

Common anchor-residual formulation under matched information

Main-text concept figure - scientific abstraction, not a layer-by-layer network diagram

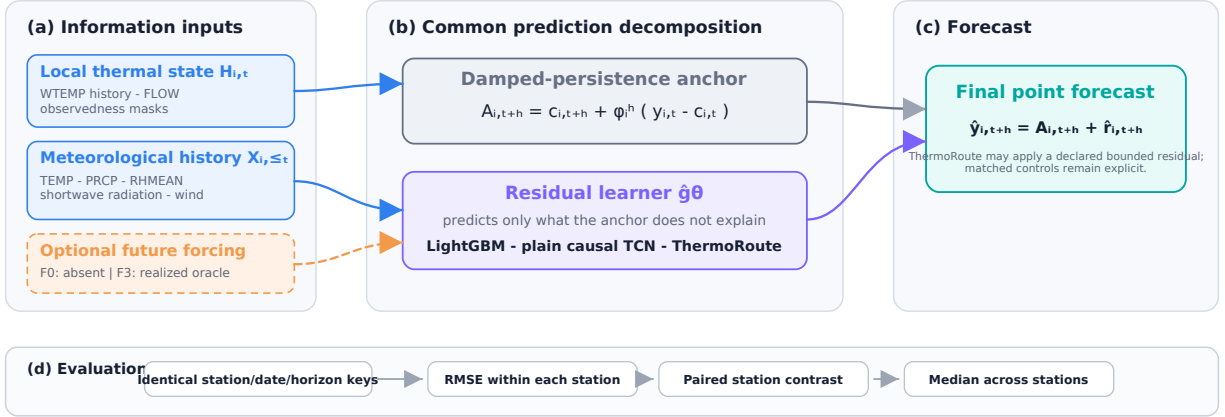


Figure 1: Common anchor-residual formulation under matched information. Historical local state and meteorological inputs feed a fixed damped-persistence anchor and a learned residual model. The final forecast is scored using a common station/date/horizon registry and a station-first paired estimator.

ThermoRoute Figure Compilation Example

1 Main-text model concept

Experimental identification of river-temperature information regimes

Core design: 2 forcing levels x 2 local-information levels x 2 geometries x 2 model classes x 3 leads = 48 matched cells

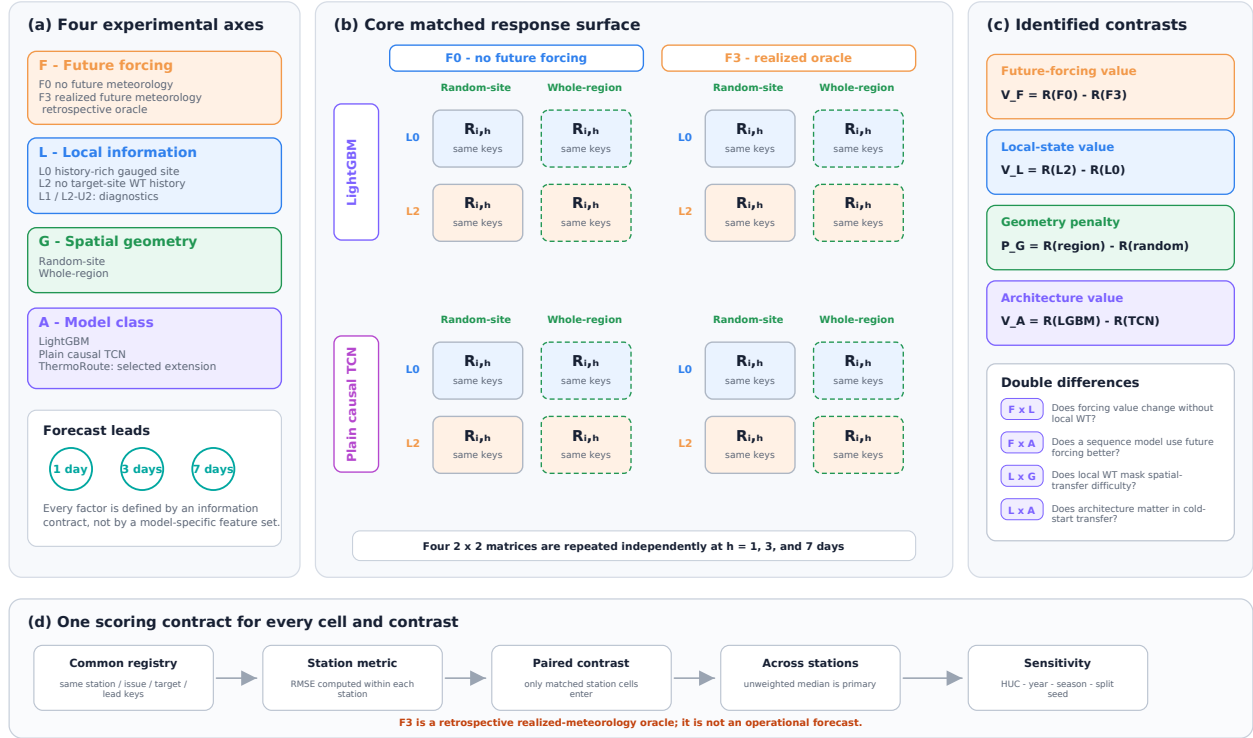


Figure 2: Experimental identification of river-temperature information regimes. The core design crosses future forcing (F), target-site local information (L), spatial geometry (G), and model class (A), repeated at 1-, 3-, and 7-day leads. Main effects and double differences are calculated from station-level RMSE values on identical forecast keys. F3 is a retrospective realized-meteorology oracle and is not an operational forecast.

2 F x L x G x A identification design

Full ThermoRoute architecture and calibration dataflow

Supporting Information figure - full scientific modules, simplified computational operators

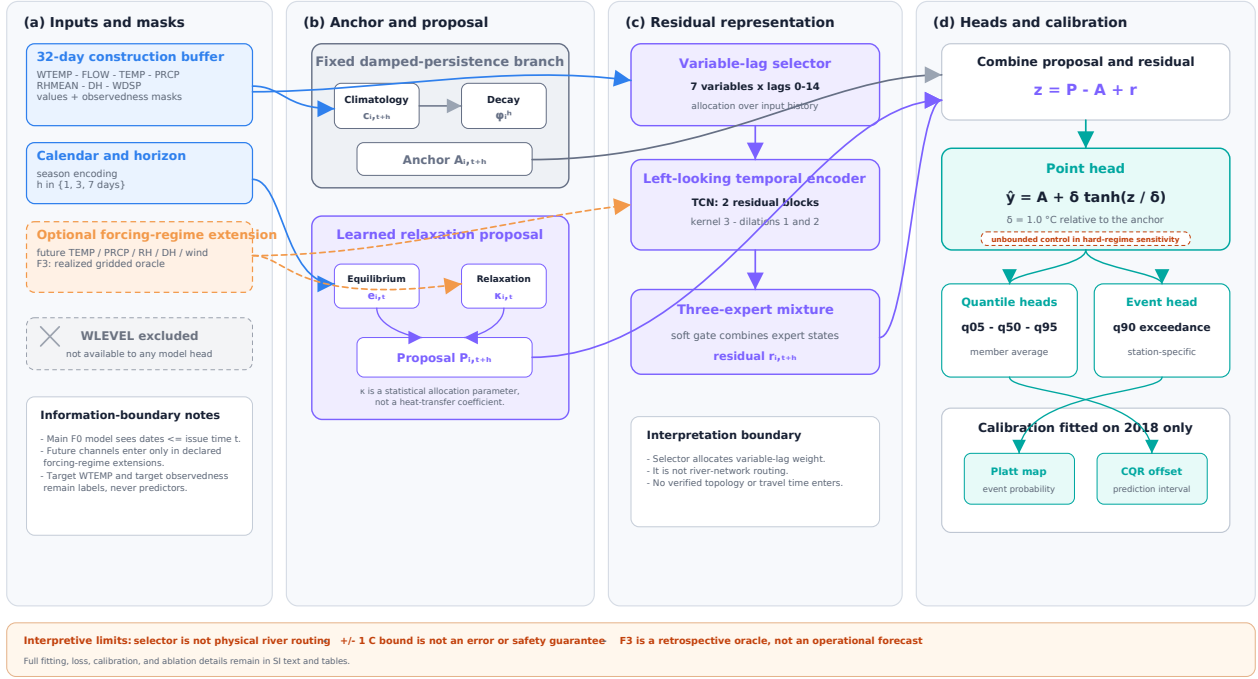


Figure 3: Full ThermoRoute architecture and calibration dataflow. The variable-lag selector is an internal allocation mechanism and does not represent physical river-network routing. The residual bound is relative to the damped-persistence anchor and is not an error or safety guarantee.

3 Supporting-Information architecture